

Product Requirements

AR Science

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Brief problem statement

The motivation for this project is to help upper elementary to middle school aged students learn about plant and animal cells and their parts. Cells can be a complicated topic for students as it is, and two-dimensional pictures don't do the best job of showing the full structure of a cell. By using augmented reality, our application will allow students to explore these cells and be able to zoom in on each of its parts through three-dimensional models. The problem we are intending to solve is the students' difficulty understanding cell structures and their functions before entering high school.

System requirements

A smartphone running on Android or IOS with a working camera is needed for the software and is developed for modern versions of IOS and Android.

Users profile

The system is intended for upper elementary to middle school aged students. The characteristics the users should possess is basic familiarity with using a smartphone, no prior knowledge of cells is needed.

List of Features

Functional requirements (user stories)

List the Priority as 1 (High Priority - Critical) to 3 (Low Priority – Would be nice if we have time)

No.	User Story Name	Description	Priority
F1	View cell	The app should allow users to pick and view a cell in AR.	1
F2	Switch cells	The user can choose to switch between different types of cells for viewing.	2
F3	Inspect cells	The app should display additional information about parts of the cell that the user indicates.	1

F4	Zoom in on cells	The app should allow users to zoom in on the cells.	1
F5	Types of cells	There should be several types of plant and animal cells to view.	2
F6	Visualize cell energy flow	The app should show how energy moves through and is used by the cell.	3
F7	View cell DNA/RNA	The app should show the cell's genetic material.	3
F8	Show cell division	The app should show the process of cell division.	3
F9	Show environmental response	The app should show the cell's response to an environmental stimulus.	3
F10	Rotate cells	The app should allow users to rotate cell models to be able to view it from different angles	1
F11	Compare cells	The app should allow users to compare different types of cells	2
F12	Tutorial	The app should provide a tutorial on how to use the app properly	1
F13	Show/hide labels	The app should allow users to either show or hide the labels identifying the cells	2
F14	AR recalibration	Lets you recalibrate the AR tracking	2

Non-Functional Requirements

Security.

- NF1: The app will not collect or store any identifiable information.

Reliability

- NF2: The system will be available during scheduled demonstrations and evaluation periods.

Usability

- NF3: The system will follow the 7 universal design principles.
- NF4: The system will meet Apple and Android accessibility standards for apps.

Cross-Platform Compatibility

- NF5: The System will work on iPhone 6s/iPhone SE or newer
- NF6: The System will work on phones that have Android 7.0 or later.

- NF7: The System will not require any additional plugins or software installations for the user.

Accuracy

- NF8: The app will contain only accurate information obtained from reliable sources.

Sponsor Requirements

I have read and approved the material in this document. If there is no external sponsor, the TA or instructor will sign it for accuracy/scope.

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Print Name	Signature	Date